

Nicholas Wright

828.775.8588 ♦ nickawright1991@gmail.com ♦ linkedin.com/in/nawrigh7 ♦ github.com/nawrigh7

EDUCATION

Bachelor of Science in Computer Science, Colorado State University *May 2026*

TECHNICAL SKILLS

- **Languages:** Java, JavaScript, Python, C, C++, HTML, CSS, PHP, SQL (MySQL, PostgreSQL, MariaDB, MongoDB), Assembly, Bash
- **Operating Systems:** Linux (Fedora, Ubuntu, Debian), macOS, Windows, Raspberry Pi
- **Frameworks & Tools:** Git, React, Spring Boot, Spark, Postman, Maven, Gradle, Make, JTB, JavaCC
- **Practices:** Agile / Scrum, Full-Stack & Back-End Development, Test-Driven Development, API Integration, Data Structures & Algorithms

PROJECTS

MiniJava to MIPS Compiler – *Colorado State University* *Spring 2026*

Java, JTB, JavaCC, Vapor / Vapor-M IR, MIPS Assembly

- Developed a full compilation pipeline in Java: MiniJava → Vapor → Vapor-M → MIPS, spanning grammar parsing, type checking, and Abstract Syntax Tree (AST) construction and traversal
- Built a static type-checking system using the visitor pattern to traverse the AST and enforce MiniJava's type rules, catching errors prior to code generation
- Designed a multi-stage intermediate representation (IR) pipeline, progressively lowering high-level language constructs into machine-level instructions across four distinct compiler passes
- Implemented register allocation and stack frame management when lowering Vapor-M to MIPS, using a linear scan algorithm and graph coloring
- Validated end-to-end correctness by testing generated MIPS output against expected program behavior, achieving a 100% pass rate across the provided test suite

Machine Learning Image Analysis – *Colorado State University* *Fall 2024*

Java, Gradle, JUnit, Machine Learning

- Designed, implemented, and trained a Perceptron from the ground up for image similarity classification
- Analyzed up to 950 images within 10ms per comparison, evaluating similarity to a reference image with 0.87 accuracy using Intersection over Union (IoU) measurement
- Implemented multiple similarity-comparison methods, including pixel-based, histogram, normalized histogram, and inverse square difference approaches
- Validated results through black-box and white-box testing with JUnit, following Test-Driven Development practices

RELEVANT TECHNICAL EXPERIENCE

IT Help Desk Support / Web Developer – *NC State University & Wright Information Systems* *2012 – 2013*

- Provided direct technical support and troubleshooting for hardware, software, and network issues, escalating complex cases to specialized teams as needed
- Developed a front-end website and implemented early security features for a client-facing e-commerce build, including remote system-monitoring software

LEADERSHIP EXPERIENCE

Executive Sous Chef – *Morrison Healthcare* *May 2023 – May 2024*

- Led and managed a team of managers and supervisors overseeing 300+ culinary staff, coordinating operations across multiple locations
- Held accountable to team members, clients, and state / third-party compliance inspections while maintaining product and service standards
- Drove a 17% reduction in costs and an 8.5% increase in revenue within the first two months in the role

Sous Chef – *Jax Fish House & Oyster Bar* *July 2021 – May 2023*

- Led a small team through all stages of service in a fast-paced, quality-driven environment, from ordering through final plating
- Built and maintained vendor relationships, expanding to new vendors and products through local outreach
- Ensured compliance with personal, company, and state standards for product quality, safety, and execution